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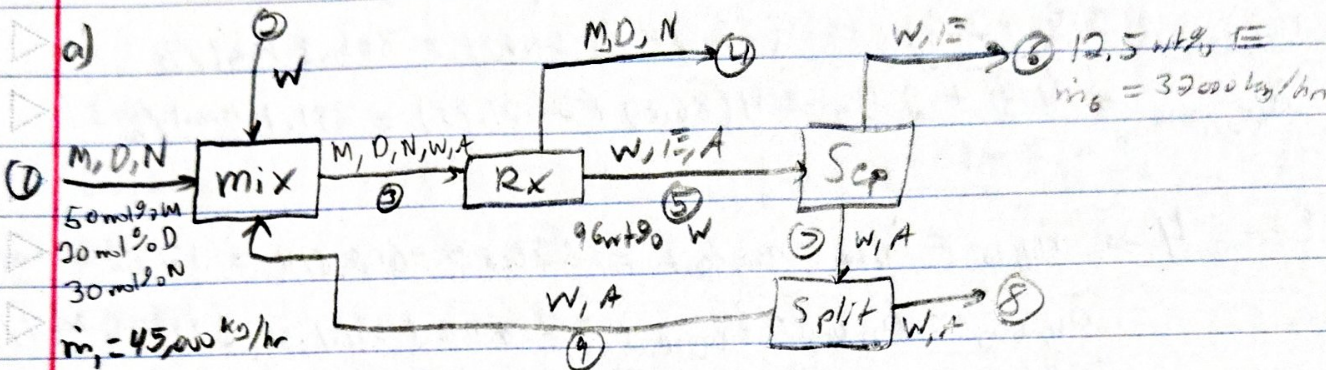
Prof. Wedgewood Homework Set 8

CHE 210

1. $6CO + 3H_2O \longrightarrow C_2H_5OH + 4CO_2$

$4CO + 2H_2O \longrightarrow CH_3COOH + 2CO_2$

a)



DoF = 0 ✓ well posed

b) $M_{CO} = 28.01$ kg/kmol $M_{CO_2} = 44.01$ kg/kmol $M_{N_2} = 28.02$ kg/kmol

$M_{H_2O} = 18.016$ kg/kmol $M_{C_2H_5OH} = 46.068$ kg/kmol $M_{CH_3COOH} = 59.044$ kg/kmol

Stream 1 → $\overline{MW}_1 = 0.5 \cdot 28.01\,kg + 0.2 \cdot 44.01\,kg + 0.3 \cdot 28.02\,kg = 31.213$ kg/kmol

$\dot{n}_1 = 45000\,kg/h \cdot \frac{1\,kmol}{31.213\,kg} = 1441.7\,kmol/h$

$\dot{n}_{1,CO} = 1441.7\,kmol/h \cdot 0.5 = 720.85\,kmol\,CO/h$ ✓

$\dot{n}_{1,CO_2} = 1441.7\,kmol/h \cdot 0.2 = 288.34\,kmol\,CO_2/h$ ✓

$\dot{n}_{1,N_2} = 1441.7\,kmol/h \cdot 0.3 = 432.51\,kmol\,N_2/h$ ✓

Reactor $\rightarrow$ $\dot{n}_{CO,react} = x_{CO}\,\dot{n}_{1,CO} = 0.8443 \cdot 720.85 \text{ kmol}$

$= 608.614 \text{ kmol/h}$

$\xi_1 = \dot{n}_E = 86.8 \text{ kmol/h}$ ✓

$6\xi_1 + 4\xi_2 = \dot{n}_{CO,react} \rightarrow \xi_2 = \dfrac{\dot{n}_{CO,react} - 6\xi_1}{4}$

$\xi_2 = \dfrac{608.614 - 6(86.8)}{4} = 21.95 \text{ kmol/h}$ ✓

$\dot{n}_{W,cons} = 3\xi_1 + 2\xi_2 = 3(86.8) + 2(21.95) = 304.3 \text{ kmol/h}$

$\dot{n}_{CO_2,prod} = 4\xi_1 + 2\xi_2 = 4(86.8) + 2(21.95) = 391.1 \text{ kmol/h}$

Stream 4 $\rightarrow$ $\dot{n}_{4,CO} = \dot{n}_{1,CO} - \dot{n}_{CO,react} = 720.85 - 608.614 = 112.24 \text{ kmol/h}$

$\dot{n}_{4,CO_2} = \dot{n}_{1,CO_2} + \dot{n}_{CO_2,prod} = 288.34 + 391.1 = 679.44 \text{ kmol/h}$

$\dot{n}_{4,N_2} = \dot{n}_{1,N_2} = 432.51 \text{ kmol/h}$

Stream 5 $\rightarrow$ $m_E = \xi_1 MW_E = (86.8 \text{ kmol/h})(46.068 \text{ kg/kmol}) = 3998.7 \text{ kg/h}$

$m_A = \xi_2 MW_A = (21.95 \text{ kmol/h})(59.044 \text{ kg/kmol}) = 1296 \text{ kg/h}$

$0.96 = \dfrac{m_{w,5}}{m_{w,5} + m_E + m_A} \rightarrow m_{w,5} = \dfrac{0.96}{0.04}(m_E + m_A) = \dfrac{0.96}{0.04}(3998.7 + 1296)$

$m_{w,5} = 127{,}073 \text{ kg/h}$

$\rightarrow \dot{n}_{w,5} = 127{,}073 \text{ kg/h} \cdot \dfrac{1 \text{ kmol}}{18.016 \text{ kg}} = 7053.3 \text{ kmol/h}$

$\dot{n}_{w,in} = \dot{n}_{w,5} + \dot{n}_{w,cons} = 7053.3 + 304.3 = 7357.6 \text{ kmol/h}$

Stream 6 $\rightarrow$ $\dot{n}_E = 86.8 \text{ kmol/h}$ $\quad \dot{n}_W = 1554 \text{ kmol/h}$

Stream 7 $\rightarrow$ $\dot{n}_W = \dot{n}_{5,W} - \dot{n}_{6,W} = 7053.3 - 1554 = 5500 \text{ kmol/h}$

$\dot{n}_A = \xi_2 = 21.95 \text{ kmol/h}$

Splitter: Stream 8 $\rightarrow \dot{n}_W = 5500 \text{ kmol/h} \cdot 0.45 = 2475 \text{ kmol/h}$

$\dot{n}_A = 21.95 \text{ kmol/h} \cdot 0.45 = 9.88 \text{ kmol/h}$

Stream 9 $\rightarrow \dot{n}_W = 5500 \text{ kmol/h} \cdot 0.55 = 3025 \text{ kmol/h}$

$\dot{n}_A = 21.95 \cdot 0.55 = 12.07 \text{ kmol/h}$

Stream 2 $\rightarrow \dot{n}_{2,W} = \dot{n}_{W,in} - \dot{n}_{9,W} = 7357.6 - 3025 = 4332.6 \text{ kmol/h}$ ✓

Stream 1: $\dot{n}_{1,CO} = 720.85 \text{ kmol/h}$ $\dot{n}_{1,CO_2} = 288.34 \text{ kmol/h}$ $\dot{n}_{1,N_2} = 432.51 \text{ kmol/h}$

Stream 2: $\dot{n}_{2,W} = 4332.6 \text{ kmol/h}$

Stream 3: $\dot{n}_{3,CO} = 720.85 \text{ kmol/h}$ $\dot{n}_{3,CO_2} = 288.34 \text{ kmol/h}$ $\dot{n}_{3,N_2} = 432.51 \text{ kmol/h}$

$\dot{n}_{3,W} = 7357.6 \text{ kmol/h}$ $\dot{n}_{3,A} = 12.07 \text{ kmol/h}$

Stream 4: $\dot{n}_{4,CO} = 112.24 \text{ kmol/h}$ $\dot{n}_{4,CO_2} = 679.44 \text{ kmol/h}$

$\dot{n}_{4,N_2} = 432.51 \text{ kmol/h}$

Stream 5: $\dot{n}_{5,W} = 7053.3 \text{ kmol/h}$ $\dot{n}_{5,E} = 86.8 \text{ kmol/h}$ $\dot{n}_{5,A} = 21.95 \text{ kmol/h}$

Stream 6: $\dot{n}_{6,W} = 1554 \text{ kmol/h}$ $\dot{n}_{6,E} = 86.8 \text{ kmol/h}$

Stream 7: $\dot{n}_{7,W} = 5500 \text{ kmol/h}$ $\dot{n}_{7,A} = 21.95 \text{ kmol/h}$

Stream 8: $\dot{n}_{8,W} = 2475 \text{ kmol/h}$ $\dot{n}_{8,A} = 9.88 \text{ kmol/h}$

Stream 9: $\dot{n}_{9,W} = 3025 \text{ kmol/h}$ $\dot{n}_{9,A} = 12.07 \text{ kmol/h}$

c) $f_{sp,W} = \frac{\dot{n}_{W,loss}}{\dot{n}_{W,in}} = \frac{304.3}{7357.6} = 0.0414$ or $\boxed{4.14\%}$ ✓

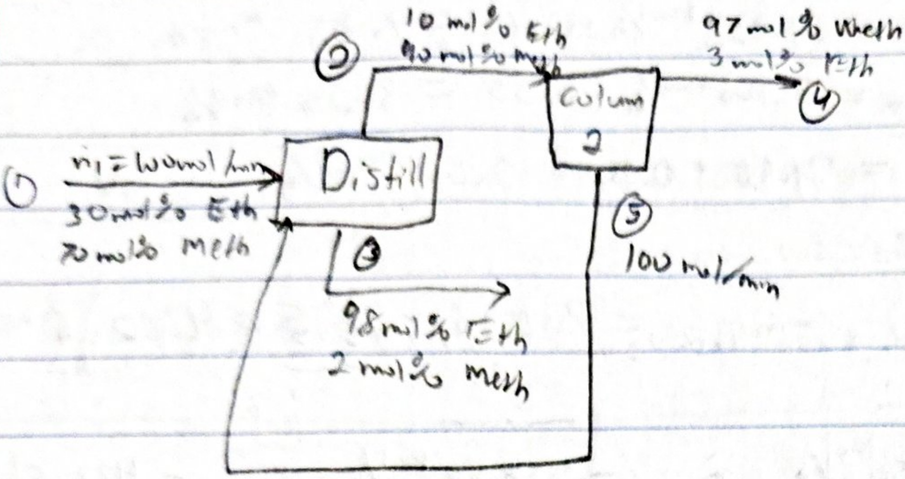
$f_{overall,W} = \frac{\dot{n}_{W,loss}}{\dot{n}_{2,W}} = \frac{304.3}{4332.6} = 0.0704$ or $\boxed{7.04\%}$ ✓

38/40

decimal difference

To increase efficiency of water use, we can reduce the amount of water that gets purged from the splitter or we can reduce the water lost in stream 6 and reroute it to recycle stream.

2.



$$\dot{n}_{in} = \dot{n}_1 + \dot{n}_5 = 100 \text{ mol/min} + 100 \text{ mol/min} = 200 \text{ mol/min}$$

$$\dot{n}_{1,E} = 100(0.3) = 30 \text{ mol/min} \qquad \dot{n}_{1,M} = 100(0.7) = 70 \text{ mol/min}$$

Column 1: $0.90\dot{n}_2 + 0.02\dot{n}_3 = 70 + 100x_{CH_4}$

$$0.90\dot{n}_2 + 0.02(200 - \dot{n}_2) = 70 + 100x_{CH_4}$$

$$0.90\dot{n}_2 + 4 - 0.02\dot{n}_2 = 70 + 100x_{CH_4}$$

$$0.88\dot{n}_2 = 66 + 100x_{CH_4}$$

$$\dot{n}_2 = \frac{66 + 100x_{CH_4}}{0.88}$$

Column 2: $\dot{n}_2 = \dot{n}_4 + 100 \rightarrow \dot{n}_4 = \dot{n}_2 - 100$

CH_4: $0.97\dot{n}_4 + 100x_{CH_4} = 0.9\dot{n}_2$

$$0.97(\dot{n}_2 - 100) + 100x_{CH_4} = 0.9\dot{n}_2$$

$$0.97\dot{n}_2 - 97 + 100x_{CH_4} = 0.9\dot{n}_2$$

$$0.07\dot{n}_2 = 97 - 100x_{CH_4}$$

$$\dot{n}_2 = \frac{97 - 100x_{CH_4}}{0.07}$$

$$\rightarrow \frac{66+100\,x_{CH_4}}{0.88} = \frac{97-100\,x_{CH_4}}{0.07}$$

$$(0.07)(66+100\,x_{CH_4}) = (0.88)(97-100\,x_{CH_4})$$

$$4.62+7\,x_{CH_4} = 85.36-88\,x_{CH_4}$$

$$95\,x_{CH_4} = 80.74 \rightarrow x_{CH_4} = 0.85$$ ✓

$$\boxed{\text{Recycle stream: } y_M = 0.85 \text{ mol/mol} \quad y_E = 0.15 \text{ mol/mol}}$$ ✓

$$\dot{n}_2 = \frac{66+100(0.85)}{0.88} = 171.59 \text{ mol/min}$$

$$\dot{n}_4 = \dot{n}_2 - 100 = 171.59 - 100$$

$$\boxed{\dot{n}_4 = 71.59 \text{ mol/min}}$$ ✓ ✓

$$\frac{CH_4 \text{ in product}}{CH_4 \text{ in fresh feed}} = \frac{0.97(71.59)}{70} = 0.992 \text{ or } \boxed{99.2\%}$$

$$\frac{x_E\,\dot{n}_3}{30} = \frac{(0.98)(28.41)}{30} = 0.928 \text{ or } \boxed{92.8\%}$$ ✓

$$\dot{n}_3 = 200 - \dot{n}_2 = 200 - 171.59 = 28.41 \text{ mol/min}$$

35/35

3. 30 mol% E 70 mol% W 1 atm

50% v and 50% liq Basis = 1 mol

$F_E = 0.3$ mol $F_W = 0.7$ mol $V = 0.5$ mol $L = 0.5$ mol

$$F_E = V \cdot y_E + L \cdot x_E$$

$$0.3 = 0.5 \cdot y_E + 0.5 \cdot x_E \quad \| \cdot 2$$

$$0.6 = y_E + x_E$$

$$\rightarrow y_E = 0.6 - x_E$$

① $85.3^\circ C \rightarrow x_E + y_E \rightarrow 0.1238 + 0.4704 = 0.5942 \approx 0.6$

$x_E = 0.1238 \quad y_E = 0.4704$

$x_W = 1 - 0.1238 = 0.8762 \quad y_W = 1 - 0.4704 = 0.5296$

Vapor $\rightarrow$ $V \cdot y_E = 0.5 \cdot 0.4704 = 0.2352$

$\frac{0.2352}{0.3} = 0.784$ or $\boxed{78.4\%}$ ✓

$100 - 78.4 = 21.6\%$

$\boxed{78.4\% \text{ ethanol and } 21.6\% \text{ water in vapor}}$ ✓

liquid $\rightarrow$ $L \cdot x_W = 0.5(0.8762) = 0.4381$

$\frac{0.4381}{0.7} = 0.626 = 62.6\%$

$100 - 62.6 = 37.4$

25/25

$\boxed{37.4\% \text{ ethanol and } 62.6\% \text{ water in liquid}}$ ✓

Separation factor $\rightarrow$ $\frac{y_E/(1-y_E)}{x_E/(1-x_E)} = \frac{0.4704/(1-0.4704)}{0.1238/(1-0.1238)} = \boxed{6.286}$

✓